

TELECOM CUSTOMER CHURN ANALYSIS FOR TELECOM INDUSTRY

Introduction:

Customer churn is one of the most significant challenges faced by the telecommunications industry. Losing existing customers directly impacts revenue, customer lifetime value, and overall business growth. This project focuses on predicting customer churn using machine learning techniques and identifying the major factors influencing customer attrition. The analysis also provides business insights and recommendations to help telecom companies improve customer retention and reduce churn.

Abstract:

The objective of this project is to develop a data-driven customer churn prediction system using Python, SQL, Machine Learning, SHAP Explainability, and Power BI. The project begins with data preprocessing, exploratory data analysis, feature engineering, and SQL-based business analysis. Multiple machine learning models including Logistic Regression, Decision Tree, and Random Forest were trained and evaluated to predict customer churn. SHAP explainability was implemented to interpret model predictions and identify the most influential features. Customers were further segmented into business categories such as High Value, Loyal, At Risk, Regular, and New Customer. Finally, an interactive Power BI dashboard was developed to visualize key business metrics, customer behavior, and churn insights, enabling informed business decision-making.

Tools Used:

- Programming Language: Python
- Database: MySQL
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SHAP, Joblib
- Machine Learning Models: Logistic Regression, Decision Tree, Random Forest
- Data Visualization: Power BI
- Development Environment: PyCharm

Steps Involved in Building the Project:

- ✓ Collected and understood the telecom customer churn dataset.
- ✓ Performed data cleaning by handling missing values, correcting data types, and preparing the dataset.
- ✓ Conducted Exploratory Data Analysis (EDA) to understand customer behavior and identify important trends.
- ✓ Imported the cleaned dataset into MySQL and performed SQL queries to generate business insights and KPIs.
- ✓ Applied feature engineering techniques including categorical encoding and dataset preparation for machine learning.
- ✓ Trained and evaluated Logistic Regression, Decision Tree, and Random Forest classification models.
- ✓ Compared model performance using Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix.
- ✓ Implemented SHAP Explainability to identify the key factors influencing customer churn predictions.
- ✓ Segmented customers into High Value, Loyal, At Risk, Regular, and New Customer categories for business analysis.
- ✓ Developed a three-page interactive Power BI dashboard presenting executive KPIs, customer behavior analysis, and customer segmentation insights.
- ✓ Generated actionable business recommendations to improve customer retention and reduce churn.

Conclusion :

This project successfully demonstrates an end-to-end Customer Churn Analysis solution for the telecom industry by integrating SQL, Python, Machine Learning, SHAP Explainability, Customer Segmentation, and Power BI. The developed predictive models effectively identify customers likely to churn, while the dashboard provides interactive business insights for decision-makers. The project highlights the practical application of data analytics and machine learning in solving real-world business problems and supports organizations in developing data-driven customer retention strategies.